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Scalable Fabrication of Intrinsically Stretchable Organic Electrochemical Transistors with Neuromorphic Functions

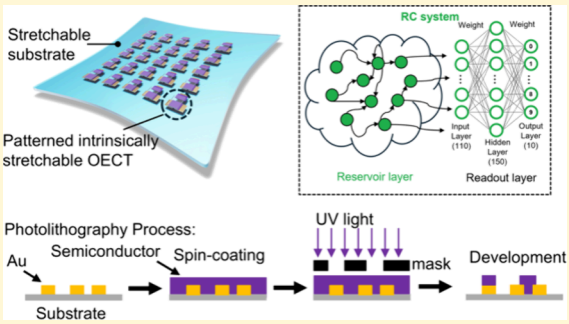
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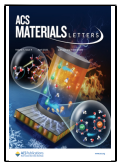
ABSTRACT: Organic electrochemical transistors (OECTs) have recently emerged as promising platforms for wearable sensors and computing devices. However, the scalable manufacturing of stretchable OECT remains a significant challenge, limiting their advancement in multicomponent sensing and massive data processing. Here, we present a photo-cross-linking strategy for the scalable fabrication of the intrinsically stretchable organic electrochemical transistors (IS-OECTs). This innovative approach allows for precise patterning of the organic semiconductor, specifically p(g2T-T). Notably, the photo-cross-linked p(g2T-T) maintains its integrity without cracking even under a 200% strain, ensuring consistent device performance under extreme strain conditions. By harnessing the nonlinear response and fading memory effect of the p(g2T-T)-based IS-OECT array, we developed a stretchable reservoir computing system that achieved impressive image recognition accuracies of 90.81% and 90.65% at 0% and 100% strains, respectively.



Organic electrochemical transistors (OECTs) are increasingly recognized as promising platforms for the development of advanced sensing and computing systems, attributed to their distinctive features such as low operation voltage, high on-site signal amplification capability, and ionic dynamic controlled channel conductivity.[1−9] Recently, there has been a prevalent focus on achieving stretchable OECTs, with the aim of fully harnessing the potential of OECT-based devices for skin-interfaced sensing and computing applications.[10−20] For instance, Cicoira et al. developed stretchable OECTs by synergistically reducing film thickness and adjusting baking temperature, optimizing them for skin-integrated pressure sensors.[17] They also achieved successful fabrication of microscale stretchable OECTs on PDMS using parylene pattern transfer and orthogonal photolithography techniques.[21] Moreover, Zhang and Khademhosseini et al. introduced an innovative hydrogel-enabled transfer printing technology for skin-integrated OECTs fabrication. This approach enabled the development of skin-attachable tattoo-OECTs, showcasing their potential for seamless wearable applications.[18] Additionally, Marks and Facchetti et al. demonstrated highly stretchable OECTs with consistent current output under biaxial stretching, utilizing a prestretched semiconductor with a honeycomb porous structure.[15] This stable response of the OECT to deformation enables reliable biosignal detection and neuromorphic applications. In a recent development, Zhang et al. reimagined the architecture of stretchable OECTs, introducing vertical intrinsically stretchable OECTs (IS-OECTs) that hold promise for skin-interface sensing and edge computing applications.[22] While significant progress has been achieved, these efforts have primarily focused on achieving stretchability of OECTs at the individual device level.[23,24] To promote the practical application of stretchable OECTs, scalable fabrication technologies are required. Zhang et al. recently introduced a six-channel inkjet printing technology for producing an array of IS-OECTs. While we appreciate the substantial progress this work represents, inkjet printing has inherent limitations like low throughput and device density.[20]

Here, we presented a photo-cross-linking approach for the high throughput and scalable fabrication of IS-OECTs. The photo-cross-linking process imparts patterning capabilities to the previously developed stretchable organic mixed ionic-electronic semiconductor/conductor (OMIEC), i.e., p(g2T-T), while maintaining its high stretchability (i.e., able to

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withstand 200% strain without developing cracks). The IS-OECT based on photo-cross-linked p(g2T-T) provided the desired electrical and mechanical characteristics with high transconductance (G_m) of 10 mS even at 200% strain. We further verified the high throughput and scalable fabrication capability of our photo-cross-linking strategy by fabricating high density OECT array (>1000 device/cm^2). To showcase the potential applications of our IS-OECT array, we created a stretchable reservoir computing (RC) system that utilized the ionic-dynamically controlled nonlinear response and fading memory properties of the IS-OECT array. This IS-OECTs-based RC system achieved notable image recognition accuracies of 90.65% at 100% strain, demonstrating its robust performance.

The IS-OECT array was designed with four stretchable components: (i) a stretchable patternable semiconductor film, (ii) stretchable Au-elastic polymer intermixing electrodes serving as both S/D/G, (iii) a stretchable ion-gel layer, and (iv) an elastic substrate (Figure 1a). p(g2T-T) was chosen as

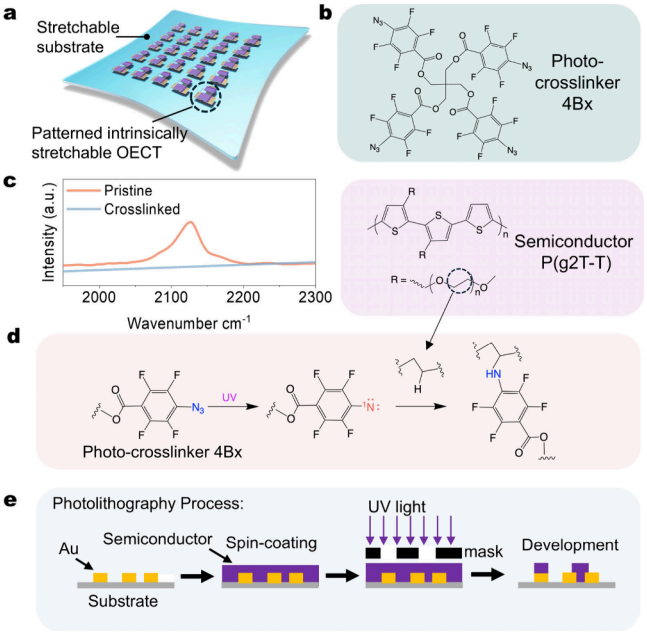


Figure 1. (a) Schematic illustration of the IS-OECT array. (b) Chemical structures of 4Bx and p(g2T-T). (c) FTIR spectra of p(g2T-T)/4Bx blended film. (d) Mechanism of the photo cross-linking reaction between p(g2T-T) and 4Bx. (e) The photolithography process of the semiconductor layer.

the channel material due to its classification as one of the typical OMIECs and its proven high stretchability.[14,25] However, the patterning of the p(g2T-T) is a challenge due to the lack of chemical orthogonality between the active polymer materials and other organic solutions used in the fabrication process.[26] To address this issue, the azide-based photo cross-linker (referred to as 4Bx) was utilized to selectively cross-link the p(g2T-T) film (Figure 1b). 4Bx was chosen as photo-cross-linker because of its good solubility in organic solvent and excellent patterning capability for electronic materials with C−H bonds.[27] Upon activation by UV light, the azide groups undergo insertion reactions with C−H bonds in the side chain of p(g2T-T) (Figure 1d), resulting in the formation of the cross-linked network in p(g2T-T) films.[28−30] This selective photo-cross-linking of the p(g2T-T) film significantly enhances the chemical resistance of the cross-linked area. Figure 1c shows the disappearance of the vibrational peak of the azide groups (2128 cm^{-1}) in the Fourier-transform infrared (FTIR) spectra of the cross-linked p(g2T-T)/4Bx film, which confirms the cross-linked reaction occurs.[31] Exposing the p(g2T-T)/4Bx film to UV light through a shadow mask can form the designed pattern after development (Figure 1e).

Previous research has highlighted the use of different types of azide-based photo-cross-linkers to enhance the patterning and insolubility of dielectric layers and organic semiconductors in the fabrication of field-effect transistors and photovoltaic devices.[28−30] However, these studies have primarily focused on patternability and device electrical performance, with little attention being given to the mechanical properties of the materials after cross-linking. Additionally, the application of azide-based cross-linking strategies in the fabrication of stretchable OECTs has not been explored. Our objective is to improve the patterning capabilities of the channel materials, such as p(g2T-T), in OECTs while preserving their high stretchability. This approach aims to achieve the scalable fabrication of stretchable OECT arrays.

Figure 2a shows the optical microscope images of the p(g2T-T) films cross-linked with various concentrations of 4Bx

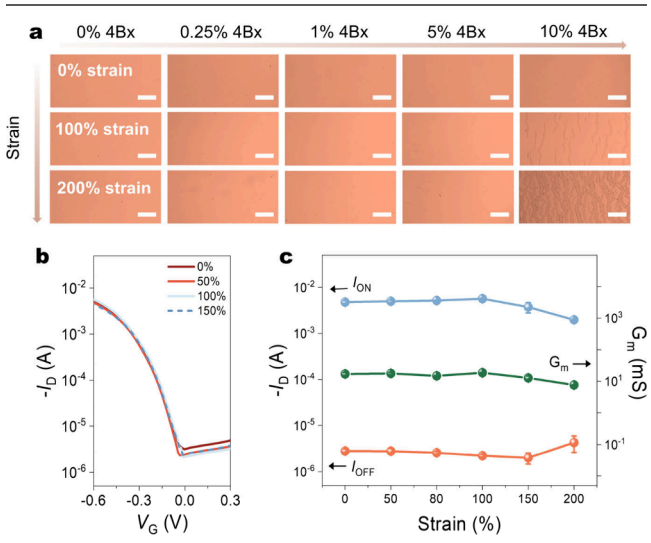


Figure 2. (a) Optical micrograph of cross-linked films with different 4Bx ratios at 0−200% strains. Scale bar: 100 μm. (b) Transfer curves (V_D = −0.6 V) of the OECT obtained from cross-linked film with 5% 4Bx under 0−150% strains in parallel to the charge-transport direction. (c) The on current (I_{ON}), off current (I_{OFF}), and G_m of cross-linked film with 5% 4Bx at 0−200% strains.

at different strains. Even under a 200% strain, the p(g2T-T) films exhibit no cracks across the range of photo-cross-linker content from 0% to 5%. Figure 2b shows the OECT performance during stretching to 150% strain using a typical OECT structure assembled by the soft contact lamination method.[14] The transfer curves remain nearly unaffected under stretching along the charge transport direction, just with a slight reduction in the peak G_m from ≈ 20 to ≈ 17.5 mS at 200% strain (Figure 2c).

Furthermore, we investigated the patterning capability of our photo-cross-linking method on p(g2T-T) with different 4Bx ratios. Figure 3a shows the optical micrograph of the patterned p(g2T-T) films. According to the images, an excessive

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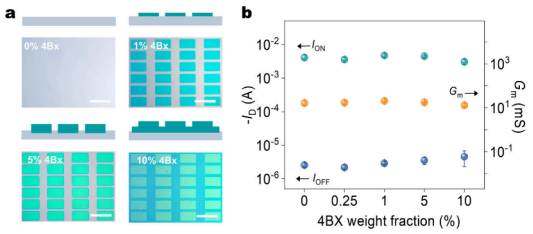


Figure 3. (a) Optical micrograph of patterned semiconductor films with different 4Bx ratios. Scale bar: 200 μm. (b) The I_{ON}, I_{OFF}, and G_m of cross-linked film with different 4Bx ratios.

concentration of 4Bx (10%) in the blend film resulted in cross-linking between the patterns. When the 4Bx concentration in the blend film is below 5% (≥1%), distinct patterns are evident. Nevertheless, it is important to mention that a high concentration of cross-linker can help prevent film degradation during the development process with organic solvents. This is supported by the observation of a thicker patterned p(g2T-T) film with 5% 4Bx compared to that with 1% 4Bx (Figure 3a).

To determine the optimal 4Bx concentration for stretchable array fabrication, we conducted additional research on the impact of the 4Bx concentration on the performance of the OECTs. During testing, an Ag/AgCl reference electrode is employed as the nonpolarizable gate.[32] The transfer curves are shown in Figure S1. Figure 3b shows the key parameters of the OECT with different ratios of 4Bx, such as I_{ON}, I_{OFF}, and G_m. Interestingly, there is no significant difference in OECT performance across cross-linker contents ranging from 0% to 5%.

Taking into account film stretchability, patterning capability, and device performance, a 5% mass fraction of the photo-cross-linker is deemed the optimal ratio for fabricating the OECT array.

To explore the stability of the cross-linked film, we utilized the 85 °C/85% RH (i.e., relative humidity) aging test to verify electrical performance of the OECT device. During 48 h 85 °C/85% RH aging (i.e., according to the empirical formula, equivalent to 1400 h at 20 °C and 30% RH), the transfer curves of p(g2T-T) and cross-linked p(g2T-T) remain nearly unaffected (Figure S2). In addition, 3600 s on/off switching (i.e., 360 cycles) and 200 repeated transfer curve tests show minor electrical performance degradation (Figures S3 and S4). The above results prove the environmental and operating stability of the OECT devices.

To demonstrate the feasibility of the photo-cross-linking strategy in the OECT array, we fabricated the high-density OECT array, as depicted in Figure S5 (1111 transistors cm^{-2}, 98% yield (tested from 50 devices)). Figure 4a depicts the schematic illustration of the high-density OECT array and device structure with coplanar S/D/G electrodes, photo-cross-linked organic semiconductors, and ion−gel layers. As shown in Figure 4b, the uniform transfer curves of 25 devices show the excellent electrical properties of the OECT array. All 25 devices show the $G_m \approx 10^3$ μS and high $I_{ON/OFF}$ ratios close to 5×10^5 (Figure 4c and 4d). In addition, batch-to-batch variation was tested from 3 batches of OECT arrays, and the result is shown in Figure S6. The photo-cross-linked OECT array exhibited minor device-to-device and batch-to-batch variation, demonstrating the effectiveness and feasibility of our azide-based photo-cross-linking method.

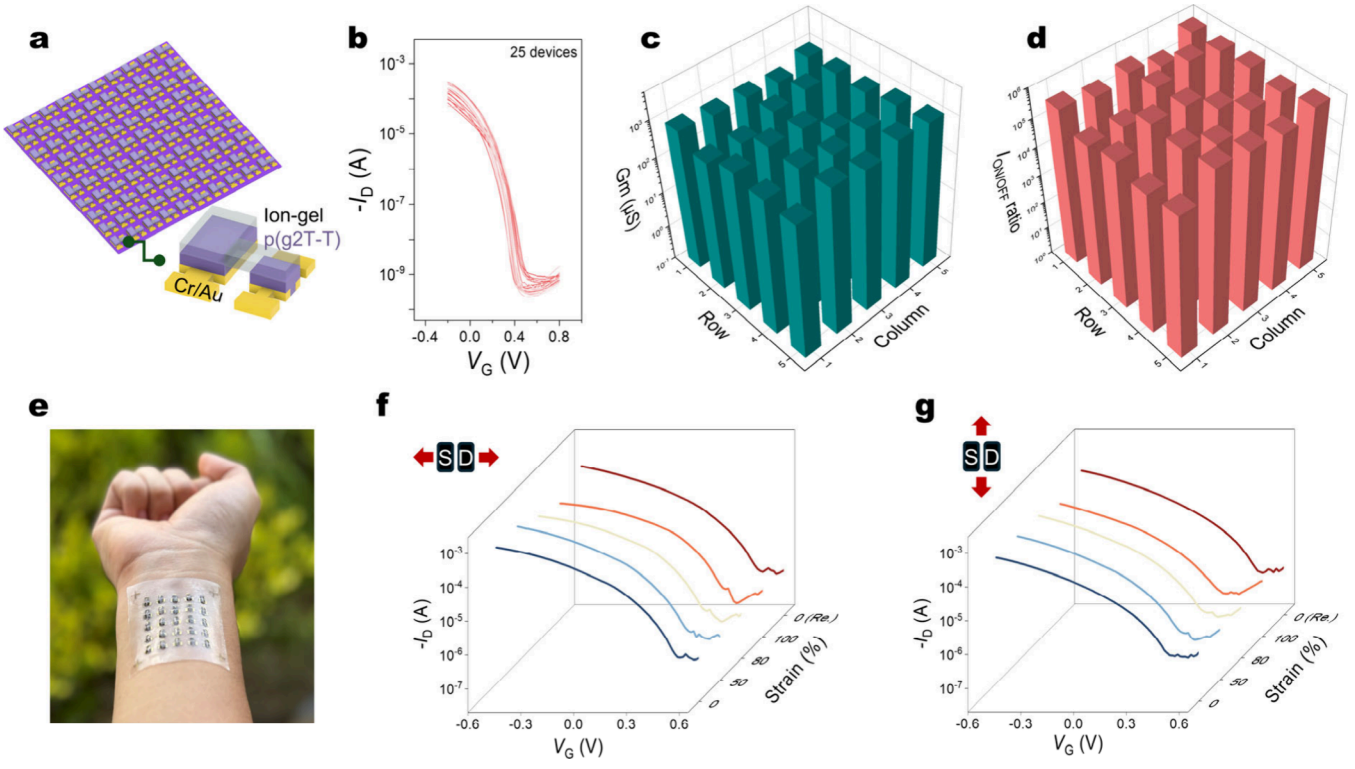


Figure 4. (a) Schematic illustration of the high-density OECT array. (b) Transfer curves (V_D = −0.1 V) of 25 transistors in the high-density OECT array. The (c) G_m and (d) $I_{ON/OFF}$ ratio distribution maps for 25 devices of the OECT array. (e) Photos of the IS-OECT array attached to the human body with skin-like conformability. Transfer curves (V_D = −0.6 V) of IS-OECTs at different stretching states, which are along two directions: (f) parallel and (g) perpendicular to the charge-transport direction.

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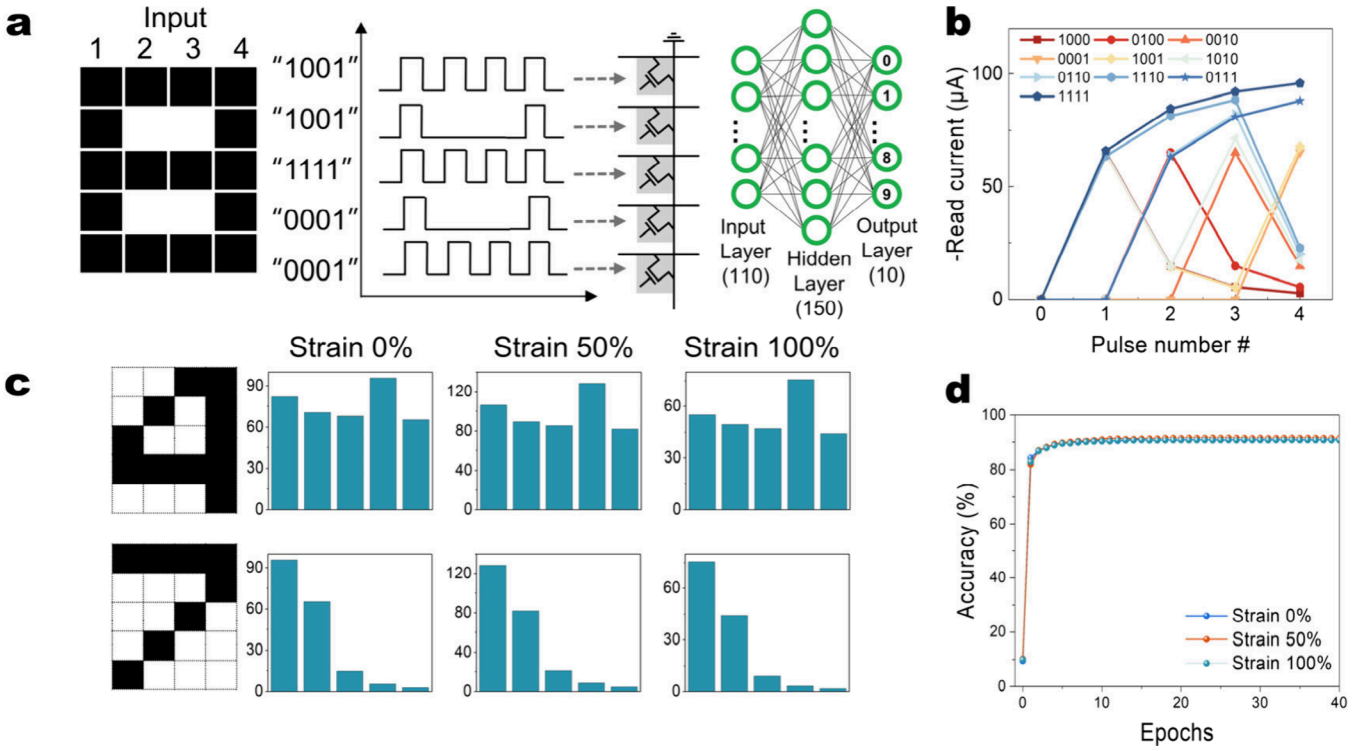


Figure 5. (a) Schematic of stretchable RC systems for image recognition. (b) Experimental read-current responses of stretchable RC systems triggered by several electrical input signals. (c) The experimental read-current responses of stretchable RC systems constituting digits "4" and "7" at 0%, 50%, and 100% strains. (d) The recognition accuracy of unstretched RC and RC systems at 50% and 100% strains along the channel length direction.

By employing this scalable fabrication approach, we successfully fabricated the IS-OECT array, as depicted in Figures 4e and S7. For the IS-OECT array, all components, including electrodes, must be stretchable. Figure S8 illustrates the resistance variation of Au-elastic polymer electrodes under different strain conditions. The intermixing layer significantly reduces the formation of large cracks in the Au electrodes during stretching, thereby preserving the high conductivity of the electrode during stretching.[33]

The basic transfer and output curves of the IS-OECT array are shown in Figure S9. For the evaluation of the flexibility of the device, the electrical performance at the bending state were explored. As shown in Figure S10, the transfer curve exhibits minimal changes under bending conditions compared with the curve measured in the flat state. The stretchability of the fully stretchable OECTs was tested by stretching the device from 0% to 100% strains. As shown in Figure 4f and 4g, the performance of the OECT exhibits only minor changes under 0−100% strain in both parallel and perpendicular directions. These slight changes observed in the transfer curves during device stretching primarily originated from the combined effects of strain-induced anisotropy in the p(g2T-T) film and device dimension (i.e., the channel length (L) and channel width (W)) changes.[14] We also conducted tests on the mechanical cyclic stability of our stretchable OECT. Even after stretching to 50% for 1000 cycles, the transfer curves exhibit only minor changes and G_m still remain 1.5 mS (Figure S11), indicating the robust mechanical stability of our fully stretchable OECTs.

In the human brain, an extensive network of synapses plays a crucial role in information transmission. Inspired by biological synapses, we developed OECT arrays leveraging ion modulation offers the potential to simulate synaptic networks for advanced information processing. The schematic diagram of biological synapse is shown in Figure S12. Initially, the presynaptic membrane is triggered by an action potential, which releases neurotransmitters. Then, the postsynaptic membrane receives these neurotransmitters, generating an excitatory/inhibitory postsynaptic current (EPSC/IPSC). Inspired by this process, the typical synaptic behavior of EPSC was simulated. Furthermore, paired-pulse facilitation (PPF) was mimicked using our device with two consecutive electrical pulses (Figure S13). In general, the formation of long-term plasticity (LTP) is regarded as a fundamental process underlying human learning and memory, emerging from short-term plasticity (STP) through repeated rehearsal. Given the importance of LTP/D for classification accuracy, we systematically tested the dependence of LTP/D curves on the amplitude and number of V_G pulses (Figure S14). The simulated synaptic behavior demonstrates the device's capability for data processing, which is highly essential for constructing neuromorphic computing systems.

To further demonstrate the potential of our IS-OECT array in neuromorphic computing, we constructed RC systems to recognize handwritten digits. Figure 5a depicts a 5 × 4 digit image "8" composed of "0" and "1", where "1" represents an input pulse (−0.5 V, 10 ms) and "0" represents no applied pulse. With four pixels as a sequence, the entire image can be encoded by five voltage sequences, which are converted into five output currents by our stretchable OECT array. RC uses the nonlinear response and decay characteristics of devices for the preprocessing of input signals, which can efficiently reduce the amount of input information and retain its properties. These sequences are converted into output currents by the IS-

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OECT array and subsequently processed through a readout layer for training and testing, thereby significantly reducing the training cost.[34] Figures 5b and S15 present the distinguishable read currents corresponding to ten sequences at different strains. Figure 5c shows the read currents for sequences corresponding to digits "4" and "7" under different strains. Due to the exceptional stretchability of OECTs, the read currents of sequences remained consistently distinguishable across various strain levels, showcasing the feasibility of our OECT array for RC applications.

Given the nonlinear response and quasi-linear conductance programmability capabilities of our stretchable OECT array, we then verified their ability for the recognition of handwritten digits in the Mixed National Institute of Standards and Technology (MNIST) data set. As shown in Figure S16, digit "8" was first binarized and then converted to a 22 × 20 pixel image. Each row of the processing image was divided into five sequences, which were fed into the RC layer of OECT array-based RC system for information reduction. Therefore, an image from MNIST can be unique encoded into 110 read currents, which was then trained and tested by the readout layer, thereby efficiently reducing the training cost. For the readout layer, the connection weights of the network were adjusted through changing the conductance of OECTs-based neurons (Figure S17). Figure 5d depicts that our RC system achieved an accuracy of over 90% after 40 training epochs, even under varying strain conditions. Figure S18 demonstrates the corresponding confusion matrix, and the recognition of each digit shows good consistency under different strains, reflecting the impressive tensile strength of our RC system. These results confirm the feasibility of the designed OECT array-based RC systems for wearable or skin-interfaced edge data processing.

In summary, we have presented a photo-cross-linking strategy for the high throughput and scalable fabrication of the IS-OECT array, endowing the previously developed intrinsically stretchable semiconductors with patterning capabilities. Utilizing the photo-cross-linking method, we fabricated the OECT array with high $I_{ON/OFF}$ ratio (5×10^5) and a density of up to 1111 transistors cm^{-2}. The electrical and mechanical performances of the IS-OECT array remain robust even under significant deformation, enduring strains of up to 100%. Additionally, the RC system based on the IS-OECT array achieved image recognition accuracies of 90.81% and 90.65% at 0% and 100% strains, respectively. Overall, this work presents a promising approach for the scalable fabrication of the IS-OECT array, ensuring reliable computing performance under varying strain conditions.

The next phase of research involves transitioning from arrays of separate stretchable OECT elements to an integrated functional circuit system. To achieve a fully integrated stretchable OECT system, several challenges must be tackled. A major issue is the low conductivity and limited stretchability of the lead electrode, which affects both the density of the device array and performance due to the voltage division effect. Additionally, as the system scales up, the complexity of interconnectivity and routing increases, necessitating optimized electrode designs and reliable contact strategies. Power consumption and cross-talk must also be carefully managed to ensure efficient signal processing, particularly for edge computing applications. The seamless integration of stretchable control peripheral circuits with the stretchable OECT array is crucial but has yet to be achieved. Another critical factor is long-term operational stability, as integrated circuits require consistent performance over extended periods that can be influenced by environmental conditions and material degradation. Collaboration among researchers with diverse backgrounds is essential for effectively tackling these challenges in future studies to promote the practical utility of stretchable OECTs.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsmaterialslett.5c00261.

Additional experimental details, transfer curves of cross-linked film, sheet resistance of stretchable Au electrodes, transfer and output curves of IS-OECT, schematic illustration of a biological synapse, PPF behavior, long-term memory, the LTP/D curves, experimental read-current responses of RC systems, schematic of simulation processes, the confusion matrix results (PDF)

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Author Contributions

[§]T.R.S. and X.L. contributed equally to this work. T.R.S.: Conducted measurements and analysis of device performances, data curation, wrote and revised manuscript. X.L.: Contributing the neuromorphic computing of handwritten image recognition, wrote and revised manuscript. P.G.: Conducted measurements and analysis of device performances. J.Y.Z.: Gave suggestions about the experiments. S.L.D.: Conceived project and supervised this project, project administration, gave suggestions about the experiments, revised manuscript. J.H.: Conceived project and supervised this project, project administration, funding acquisition, gave suggestions about the experiments, revised manuscript.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Rivnay, J.; Inal, S.; Salleo, A.; Owens, R. M.; Berggren, M.; Malliaras, G. G. Organic electrochemical transistors. *Nat. Rev. Mater.* **2018**, *3* (2), 17086.
(2) Qiu, J.; Cao, J.; Liu, X.; et al. A Flexible Organic Electrochemical Synaptic Transistor With Dopamine-Mediated Plasticity. *IEEE Electron Device Lett.* **2023**, *44* (1), 176−179.
(3) Sophocleous, M.; Contat-Rodrigo, L.; Garcia-Breijo, E.; et al. Organic Electrochemical Transistors as an Emerging Platform for Bio-Sensing Applications: A Review. *IEEE Sens. J.* **2021**, *21* (4), 3977−4006.
(4) Cucchi, M.; Gruener, C.; Petrauskas, L.; et al. Reservoir computing with biocompatible organic electrochemical networks for brain-inspired biosignal classification. *Sci. Adv.* **2021**, *7* (34), No. eabh0693.
(5) Wang, X.; Meng, X.; Zhu, Y.; et al. A sub-1V, microwatt power-consumption iontronic pressure sensor based on organic electrochemical transistors. *IEEE Electron Device Lett.* **2021**, *42* (1), 46−49.
(6) He, Y.; Zhu, L.; Zhu, Y.; Chen, C.; Jiang, S.; Liu, R.; Shi, Y.; Wan, Q. Recent Progress on Emerging Transistor-Based Neuromorphic Devices. *Adv. Intell. Syst.* **2021**, *3* (7), 2000210.
(7) Flagg, L. Q.; Bischak, C. G.; Quezada, R. J.; et al. P-Type Electrochemical Doping Can Occur by Cation Expulsion in a High-Performing Polymer for Organic Electrochemical Transistors. *ACS Mater. Lett.* **2020**, *2* (3), 254−260.
(8) Tropp, J.; Meli, D.; Wu, R.; et al. Revealing the Impact of Molecular Weight on Mixed Conduction in Glycolated Polythiophenes through Electrolyte Choice. *ACS Mater. Lett.* **2023**, *5* (5), 1367−1375.
(9) Song, J.; Li, L.; Wong, W.-Y.; et al. Organic Mixed Conductors in Electrochemical Transistors for Bioelectronic Applications. *Accounts Mater. Res.* **2024**, *5* (9), 1036−1047.
(10) Yang, A.; Song, J.; Liu, H.; Zhao, Z.; Li, L.; Yan, F. Wearable Organic Electrochemical Transistor Array for Skin-Surface Electrocardiogram Mapping Above a Human Heart. *Adv. Funct. Mater.* **2023**, *33* (17), 2215037.
(11) Jo, Y. J.; Ok, J.; Kim, S. Y.; Kim, T.-i. Stretchable and Soft Organic−Ionic Devices for Body-Integrated Electronic Systems. *Adv. Mater. Technol.* **2022**, *7* (2), 2001273.
(12) Dai, S.; Dai, Y.; Zhao, Z.; et al. Intrinsically stretchable neuromorphic devices for on-body processing of health data with artificial intelligence. *Matter* **2022**, *5* (10), 3375−3390.
(13) Wang, Y.; Liu, D.; Bai, J.; Tian, X.; Lai, P. T.; Zhang, S. Stretchable, enhancement-mode PEDOT:PSS organic electrochemical transistors. *Appl. Phys. Lett.* **2024**, *125* (14), 141901.
(14) Dai, Y.; Dai, S.; Li, N.; Li, Y.; Moser, M.; Strzalka, J.; Prominski, A.; Liu, Y.; Zhang, Q.; Li, S.; et al. Stretchable Redox-Active Semiconducting Polymers for High-Performance Organic Electrochemical Transistors. *Adv. Mater.* **2022**, *34* (23), No. e2201178.
(15) Chen, J.; Huang, W.; Zheng, D.; et al. Highly stretchable organic electrochemical transistors with strain-resistant performance. *Nat. Mater.* **2022**, *21* (5), 564−571.
(16) Peng, Y.; Gao, L.; Liu, C.; et al. Stretchable organic electrochemical transistors via three-dimensional porous elastic semiconducting films for artificial synaptic applications. *Nano Res.* **2023**, *16* (7), 10206−10214.
(17) Zhang, S.; Li, Y.; Tomasello, G.; Anthonisen, M.; Li, X.; Mazzeo, M.; Genco, A.; Grutter, P.; Cicoira, F. Tuning the Electromechanical Properties of PEDOT:PSS Films for Stretchable Transistors And Pressure Sensors. *Adv. Electron. Mater.* **2019**, *5* (6), 1900191.
(18) Zhang, S.; Ling, H.; Chen, Y.; Cui, Q.; Ni, J.; Wang, X.; Hartel, M. C.; Meng, X.; Lee, K.; Lee, J.; et al. Hydrogel-Enabled Transfer-Printing of Conducting Polymer Films for Soft Organic Bioelectronics. *Adv. Funct. Mater.* **2020**, *30* (6), 1906016.
(19) Zhang, X.; Wang, B.; Huang, L.; Huang, W.; Wang, Z.; Zhu, W.; Chen, Y.; Mao, Y.; Facchetti, A.; Marks, T. J. Breath figure-derived porous semiconducting films for organic electronics. *Sci. Adv.* **2020**, *6* (13), No. eaaz1042.
(20) Liu, D.; Tian, X.; Bai, J.; et al. A wearable in-sensor computing platform based on stretchable organic electrochemical transistors. *Nat. Electron.* **2024**, *7* (12), 1176−1185.
(21) Zhang, S.; Hubis, E.; Tomasello, G.; et al. Patterning of Stretchable Organic Electrochemical Transistors. *Chem. Mater.* **2017**, *29* (7), 3126−3132.
(22) Dai, S.; Zhang, X.; Liu, X.; Tian, X.; Cui, B.; Pang, I.; Luo, H.; Liu, D.; He, X.; Chen, X.; et al. Vertical-Structure Overcomes the Strain Limit of Stretchable Organic Electrochemical Transistors. *Adv. Mater.* **2025**, *37* (4), 2413951.
(23) Gu, P.; Lu, L.; Yang, X.; Hu, Z.; Zhang, X.; Sun, Z.; Liang, X.; Liu, M.; Sun, Q.; Huang, J.; Zu, G. Highly Stretchable Semiconducting Aerogel Films for High-Performance Flexible Electronics. *Adv. Funct. Mater.* **2024**, *34* (33), 2400589.
(24) Hu, Z.; Gu, P.; Yang, X.; Sun, Z.; Lu, L.; Liang, X.; Zhang, X.; Deng, Z.; Liu, M.; Zu, G.; Huang, J. Nanoporous Conjugated Polymer Aerogel Films for High-Performance Electrochemical Transistors. *Adv. Funct. Mater.* **2024**, *34* (52), 2410788.
(25) Nielsen, C. B.; Giovannitti, A.; Sbircea, D. T.; et al. Molecular Design of Semiconducting Polymers for High-Performance Organic Electrochemical Transistors. *J. Am. Chem. Soc.* **2016**, *138* (32), 10252−9.
(26) Guo, P.; Zhang, J.; Pu, H.; et al. Wafer-scale photolithographic fabrication of organic synaptic transistor arrays. *Device* **2024**, *2* (7), 100409.
(27) Kim, M. J.; Lee, M.; Min, H.; Kim, S.; Yang, J.; Kweon, H.; Lee, W.; Kim, D. H.; Choi, J.-H.; Ryu, D. Y.; et al. Universal three-dimensional crosslinker for all-photopatterned electronics. *Nat. Commun.* **2020**, *11* (1), 1520.
(28) Lee, J.; Hassan, S. Z.; Lee, S.; Sim, H. R.; Chung, D. S. Azide-functionalized ligand enabling organic-inorganic hybrid dielectric for high-performance solution-processed oxide transistors. *Nat. Commun.* **2022**, *13* (1), 7021.
(29) Lee, M.; Choi, B.-i.; Ahn, P.; et al. Low-Voltage Organic Transistors with Carrier Mobilities over 10 cm2V−2s−1 Using Six-Branched Organic Azide. *Chem. Mater.* **2022**, *34* (23), 10409−10423.
(30) Dahlstrom, S.; Wilken, S.; Zhang, Y.; et al. Cross-Linking of Doped Organic Semiconductor Interlayers for Organic Solar Cells: Potential and Challenges. *ACS Appl. Energy Mater.* **2021**, *4* (12), 14458−14466.
(31) Wu, Y.; Dai, S.; Liu, X.; Guo, P.; Zhang, J.; Sun, T.; Guo, Z.; Xu, Y.; Liang, H.; Xiong, L.; et al. Optical Microlithography of Perovskite Quantum Dots/Organic Semiconductor Heterojunctions for Neuromorphic Photosensors. *Adv. Funct. Mater.* **2024**, *34* (23), 2315175.
(32) Inal, S.; Malliaras, G. G.; Rivnay, J. Benchmarking organic mixed conductors for transistors. *Nat. Commun.* **2017**, *8* (1), 1767.
(33) Kim, M. H.; Jeong, M. W.; Kim, J. S.; Nam, T. U.; Vo, N. T. P.; Jin, L.; Lee, T. I.; Oh, J. Y. Mechanically robust stretchable semiconductor metallization for skin-inspired organic transistors. *Sci. Adv.* **2022**, *8* (51), No. eade2988.
(34) Liu, X.; Dai, S.; Zhao, W.; Zhang, J.; Guo, Z.; Wu, Y.; Xu, Y.; Sun, T.; Li, L.; Guo, P.; et al. All-Photolithography Fabrication of Ion-Gated Flexible Organic Transistor Array for Multimode Neuromorphic Computing. *Adv. Mater.* **2024**, *36* (21), 2312473.

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